Paulette Koronkevich

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Education

Current **University of British Columbia**, Vancouver, BC, Canada
Software Practices Lab
PhD in Computer Science

Apr. 2026 **CIRTL Practitioner**
Achieved through the Teaching Practicum program

Dec. 2024 **CIRTL Associate**
Achieved through the 8-week Foundation of Pedagogy course

Aug. 2021 **University of British Columbia**, Vancouver, BC, Canada
Software Practices Lab
MSc in Computer Science

May 2019 **Indiana University**, Bloomington, Indiana, USA
School of Informatics, Computing, and Engineering
Bachelor's of Science in Computer Science
MINORS: Mathematics, Slavic Studies

Thesis

2021 *ANF Preserves Dependent Types up to Extensional Equality*
Paulette Koronkevich
UBC Electronic Theses and Dissertations (ETDs) 2008+
[DOI]

Publications

2026 *Regions as Continuation Marks*
Paulette Koronkevich, William J. Bowman
Scheme Workshop
[DOI]

2026 *Building Teaching Skills and Community for Early-Career Computer Science Academics: Reflections from a Departmental Program*
Paulette Koronkevich, Gautam Kirshan Luhana, Meghan Allen
Western Canadian Conference on Computing Education
[DOI]

2025 *Type Universes as Kripke Worlds*
Paulette Koronkevich, William J. Bowman
ACM International Conference on Functional Programming (ICFP)
[DOI]

2025 *Type-Preserving Flat Closure Optimization*
Adam T. Geller, Sean Bocirnea, Chester Gould, **Paulette Koronkevich**, William J. Bowman
ACM Object-oriented Programming, Systems, Languages and Applications (OOPSLA)
[DOI]

2022 *ANF Preserves Dependent Types up to Extensional Equality*
Paulette Koronkevich, Ramon Rakow, Amal Ahmed, William J. Bowman
Journal of Functional Programming
[DOI]

2021 *PLIERS: A Process that Integrates User-Centered Methods into Programming Language Design*
Michael Coblenz, Gauri Kambhatla, **Paulette Koronkevich**, Jenna L. Wise, Celeste Barnaby, Joshua Sunshine, Jonathan Aldrich, and Brad A. Myers.
ACM Transactions on Computer-Human Interaction (TOCHI)
[DOI]

2020 *Obsidian: Typestate and Assets for Safer Blockchain Programming*
Michael Coblenz, Reed Oei, Tyler Etzel, **Paulette Koronkevich**, Miles Baker, Yannick Bloem, Brad A. Myers, Joshua Sunshine, and Jonathan Aldrich.
ACM Transactions on Programming Languages and Systems (TOPLAS)
[DOI]

Presentations

2026 ***Paradoxes and Proofs with Pointers***
Talk Abstract at *Women in Logic (WiL)* associated with *Federated Logic Conference (ACM/IEEE FLOC 2026)*.

2025 ***Type Universes as Kripke Worlds: Memory Management Edition***
Invited Talk at *International Workshop on Aliasing, Capabilities, and Ownership (IWACO)* associated with *Systems, Programming, Languages, and Applications: Software for Humanity conference (ACM SPLASH 2025)*.

2022-23,25 ***Making the Most out of Conferences***
Invited Talk at *Programming Languages Mentoring Workshop* associated with *International Conference on Functional Programming (ACM ICFP 2022, 2023, 2025)*.

2023 ***One Weird Trick to Untie Landin's Knot***
Invited Talk at *PL Wonks Seminar* at Indiana University.

2023 ***Control in Dependent Types***
Joint Invited Talk at *Shonan Meeting 203: Effect Handlers and General-Purpose Languages*.

2023 ***One Weird Trick to Untie Landin's Knot***
Talk Abstract at *Higher-Order Programming with Effects (HOPE)* associated with *International Conference on Functional Programming (ACM ICFP 2023)*.

2022 ***Dependent-type-preserving Memory Allocation***
Second place in the Student Research Competition at *Principles of Programming Languages (ACM POPL 2022)*.

2021 ***The ANF Translation Preserves Dependent Types up to Extensional Equality***
Invited Talk at *Languages, Systems, and Data (LSD) Seminar* at UC Santa Cruz.

2019 ***Reigniting Fuse, an Online Partial Evaluator for Scheme***
Talk Abstract at *Scheme Workshop* associated with *International Conference on Functional Programming (ACM ICFP 2019)*.

2018 ***Obsidian in the Rough: A Case Study Evaluation of a New Blockchain Programming Language***
Second place in the Student Research Competition at *Systems, Programming, Languages, and Applications: Software for Humanity (ACM SPLASH 2018)*.

Honors & Awards

May 2026 Incredible Instructor Award
May 2024 Student Community Building Award & Graduate Teaching Assistant Award
May 2023 Student Community Building Award & Graduate Teaching Assistant Award
Dec 2023 Faculty of Science Service Award
May 2022 Maria Klawe Service Award & Graduate Teaching Assistant Award
May 2021 Student Service Award
2022 Second place at Principles of Programming Languages (ACM POPL 2022) Student Research Competition
2019-2021 UBC CS Merit Scholarship
2019 NSF Graduate Research Fellowship *(but unable to receive funding at Canadian institution)*

Work Experience

Summ. 2019 **Galois Inc**, Portland, Oregon
Software Engineering/Research Intern

- Built a system to generate tests for an API based on a high-level specfication.
- Developed a prototype of a typechecker for a refinement type system.

2016-2019 **Indiana University**, Bloomington, Indiana
IU Peer Coach

- Assisted all students with registering, scheduling, and planning classes.
- Held 8-10 appointment and drop-in hours to provide support for students.
- Organized degree-planning workshops to prepare incoming students.

Summ. 2018 **Carnegie Mellon University**, Pittsburgh, Pennsylvania
Undergraduate Research Assistant

- Conducted a case study evaluation of a new blockchain programming language.
- Contributed a new feature to the language based on the case studies.
- Wrote up and presented at a student research competition.

Summ. 2017 **JPMorgan Chase & Co**, Chicago, Illinois
Tech Analyst Intern

- Learned about various software development tools and the SDLC.
- Refactored an existing application to prepare it for cloud deployment.
- Recreated the user interface into a portable widget.

Teaching (as instructor of record)

Fall 2026 CPSC 311 - Definition of Programming Languages (95 students)
Spring 2025 CPSC 411 - Compilers (44 students)
Summ. 2023 CPSC 110 - Computation, Programs, and Programming (100 students)

Teaching (as teaching assistant)

Spring 2026 CPSC 203 - Programming, problem solving, and algorithms (CS non-major course)
Spring 2023 CPSC 411 - Compilers
Spring 2022 CPSC 411 - Compilers
Fall 2021 CPSC 311 - Definition of Programming Languages
Fall 2020 CPSC 509 - Programming Languages (graduate course)
Spring 2019 CSCI-C 211 - Introduction to Computer Science, creating new labs.
Fall 2018 CSCI-C 311 - Programming Languages & EDUC-X 152 Mapping Your Future
Spring 2018 CSCI-B 490 - Dependent Types
Fall 2017 CSCI-C 311 - Programming Languages
Spring 2017 CSCI-C 211 - Introduction to Computer Science, making instructional videos.
Fall 2016 CSCI-C 211 - Introduction to Computer Science
Spring 2016 CSCI-C 211 - Introduction to Computer Science

Service

2027 Program Committee Member for the 41st European Conference on Object-Oriented Programming (ECOOP 2027).
2026 International Conference on Functional Programming (ICFP) Student Volunteer Co-Chair.
2026 Program Committee Member for the Eleventh Workshop on Mathematically Structured Functional Programming (MSFP 2026).
2025 Program Committee Member for the Scheme Workshop 2025.
2022-2023 Served on Graduate Affairs Committee for the Computer Science Department at UBC.
2022 Member of the European Conference on Object-Oriented Programming (ECOOP) Artifact Evaluation Committee.
2020-2022 President of UBC Computer Science Graduate Student Association (CSGSA).
2019-2021 Events director for UBC Women in Computer Science (WiCS).
2018-2024 Student Volunteer at International Conference on Functional Programming (ICFP) 2018-2023, Programming Language Design and Implementation (PLDI) 2019-2020, and Principle of Programming Languages (POPL) 2020-2022,2024.
2019 VP Social for UBC Computer Science Graduate Student Association.
2018 Organizational Committee for Opportunities for Undergraduate Research in Computer Science (OurCS), held at Indiana University in 2018.
Also served on a student research panel.